Roll No.: | | | | | | | | | | | |

Test Date: 18-01-2026

Aakash
Medical | IIT-JEE | Foundations

E
Booklet Code

AR-008



AIATS
ALL INDIA AAKASH TEST SERIES

Vivek Krishna

for

Medical Entrance Exam - 2026

National Eligibility-cum-Entrance Test (NEET)

TEST No. 4

(XII Passed Students)

INSTRUCTIONS FOR CANDIDATES

1. Read each question carefully.
2. It is mandatory to use Blue/Black Ball Point Pen to darken the appropriate circle in the answer sheet.
3. Mark should be dark and should completely fill the circle.
4. Rough work must not be done on the answer sheet.
5. Do not use white-fluid or any other rubbing material on answer sheet. No change in the answer once marked is allowed.
6. Student cannot use log tables and calculators or any other material in the examination hall.
7. Before attempting the question paper, student should ensure that the test paper contains all pages and no page is missing.
8. Before handing over the answer sheet to the invigilator, candidate should check that Roll No. and Centre Code have been filled and marked correctly.
9. Immediately after the prescribed examination time is over, the answer sheet to be returned to the invigilator.
10. (i) The question paper consists of 180 compulsory questions (45 questions each in Physics and Chemistry and 90 questions in Biology (Botany & Zoology)). The total duration of the examination will be 180 minutes (3 Hrs). (ii) Each correct answer carries four marks. One mark will be deducted for each incorrect answer from the total score.

Note : It is compulsory to fill **Roll No.** and **Test Booklet Code** on answer sheet, otherwise your answer sheet will not be considered.

Test No. 4

TOPICS OF THE TEST

Physics	Thermal Properties of Matter, Thermodynamics, Kinetic Theory, Oscillations, Waves
Chemistry	Solutions, Chemical Kinetics, Organic Chemistry- Some Basic Principles and Techniques
Botany	Photosynthesis in Higher plants, Respiration in plants, Plant Growth and Development
Zoology	Chemical Coordination & Integration, Animal Kingdom, Structural Organisation in Animals– Animal Morphology (Cockroach & Frog)

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MM : 720

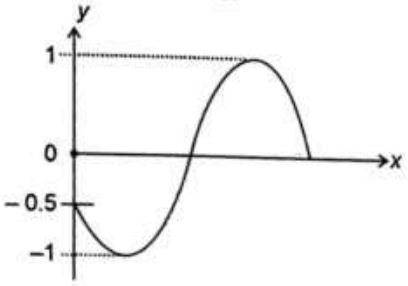
TEST - 4

Time : 3 Hrs.

[PHYSICS]

Choose the correct answer :

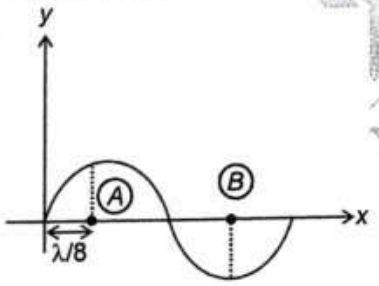
1. The snapshot of a harmonic wave travelling along positive x-axis, as shown in the figure, is at $t = 0$. Equation of this wave will be given by [All symbols have their usual meaning]



(1) $y = \sin\left(\omega t + kx + \frac{\pi}{6}\right)$ (2) $y = \sin\left(kx + \omega t - \frac{\pi}{6}\right)$

(3) $y = \sin\left(\omega t - kx + \frac{\pi}{6}\right)$ (4) $y = \sin\left(\omega t - kx - \frac{\pi}{6}\right)$

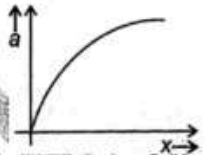
2. Snapshot of a wave travelling in a medium at any instant is shown below. The phase difference between the particles A and B of the medium located as shown will be (where λ represents the wavelength of wave)



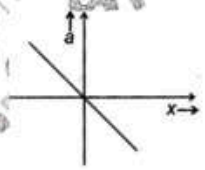
(1) $\frac{5\pi}{4}$ (2) $\frac{2\pi}{5}$

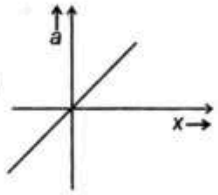
(3) $\frac{7\pi}{3}$ (4) $\frac{5\pi}{6}$

3. For a particle executing SHM, which of the following graph correctly represents the variation of acceleration (a) with its position (x) from mean position?

(1) 

(2) 

(3) 

(4) 

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Space for Rough Work

4. A particle executes SHM with frequency 4 Hz. The frequency with which its kinetic energy oscillates will be

(1) 4 Hz (2) 2 Hz

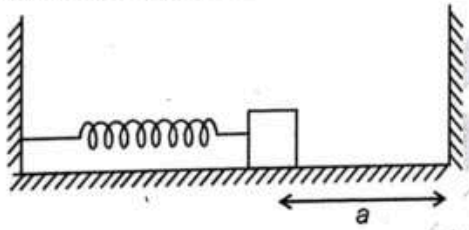
(3) 6 Hz (4) 8 Hz

5. Time period of a simple pendulum oscillating inside a lift, when lift moves downward with constant velocity v, is T. Time period of this pendulum when lift moves upward with constant acceleration $\frac{g}{4}$, will be

(1) $\frac{\sqrt{3}T}{2}$ (2) $\frac{2T}{\sqrt{5}}$

(3) $\frac{2T}{\sqrt{3}}$ (4) T

6. One end of an ideal spring of force constant k is fixed to a vertical wall and its other end is connected to a block of mass m resting on smooth horizontal surface as shown (in natural length). The spring is now compressed by a distance $2a$ and released. The time after which block will strike front wall is



(1) $2\pi\sqrt{\frac{m}{k}}$ (2) $\frac{7\pi}{6}\sqrt{\frac{m}{k}}$

(3) $\frac{3\pi}{2}\sqrt{\frac{m}{k}}$ (4) $\frac{2\pi}{3}\sqrt{\frac{m}{k}}$

7. Consider the following statements out of which one is labelled as **Assertion (A)** and other as **Reason (R)**.

Assertion (A): If in any thermodynamic gaseous process, initial volume of gas is equal to final volume of gas, then work done by gas in the process must be zero.

Reason (R): Work done is a point function.

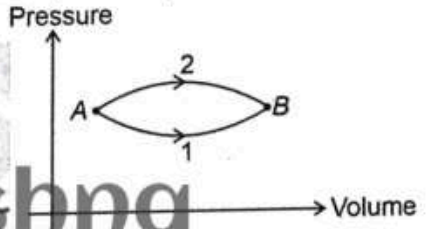
(1) Both (A) and (R) are false

(2) (A) is true but (R) is false

(3) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

8. The figure shows two paths by which a thermodynamic gaseous system is taken from state A to another state B. The ratio of molar specific heats for path 1 to path 2 will be



(1) Greater than one (2) Less than one

(3) Equal to one (4) Equal to 4/3

9. In an adiabatic thermodynamic gaseous process, the pressure P and temperature T of an ideal diatomic gas are related by relation, $P \propto T^{\alpha}$. Value of α is equal to

(1) 1.67 (2) 0.6

(3) 3.5 (4) 0.4

10. A Carnot engine works between temperature limits of 600 K and 300 K. Efficiency of the engine will be

(1) 50% (2) 70%

(3) 20% (4) 80%

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11. Consider the following statements and choose the correct option.

Statement A: Root mean square speed of molecules of an ideal gas depends on the absolute temperature of the gas.

Statement B: If heat is being extracted from a thermodynamical gaseous system, then temperature of gas must decrease.

(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Statement A is correct while statement B is incorrect
(4) Statement A is incorrect while statement B is correct

12. In the formula of mean free path $\lambda = \frac{1}{\sqrt{2}n\pi d^2}$, d represents

(1) Length of the container
(2) Diameter of molecule of gas
(3) Density of gas
(4) Distance between two collisions

13. If f denotes the degree of freedom of an ideal gas, then the ratio $\frac{C_P}{C_V}$ will be equal to (Here C_P and C_V has their usual meaning)

(1) $1+\frac{f}{2}$ (2) $\frac{2}{f}$
(3) $1+\frac{2}{f}$ (4) $1-\frac{2}{f}$

14. An ideal gas is found to obey the law, P^2V = constant. The initial temperature and volume of gas are T_0 and V_0. If the gas expands to volume $3V_0$, then final temperature of gas becomes

(1) $\sqrt{3}T_0$ (2) $\sqrt{2}T_0$
(3) $\frac{T_0}{\sqrt{3}}$ (4) $2T_0$

15. The rms speeds of the molecules of hydrogen and oxygen at same temperature are v_1 and v_2 respectively, then correct relation between v_1 and v_2 will be

(1) $v_1 > v_2$ (2) $v_1 = v_2$
(3) $v_1 < v_2$ (4) $v_1 = 2v_2$

16. If two moles of helium gas are mixed with three moles of hydrogen gas, then molar specific heat of the mixture at constant volume will be (R is universal gas constant)

(1) $1.2R$ (2) $3.4R$
(3) $7.6R$ (4) $2.1R$

17. Consider the following statements out of which one is labelled as **Assertion (A)** and other as **Reason (R)**.

Assertion (A): When a metal rod lying freely is heated, no thermal stress is developed in it.

Reason (R): On heating, the length of metal rod increases.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) Both (A) and (R) are false
(4) (A) is true but (R) is false

18. A hot body cools (as per Newton's law of cooling) from 85°C to 75°C in time t_1, from 75°C to 65°C in time t_2 and from 65°C to 55°C in time t_3, then correct relation between t_1, t_2 and t_3 will be

(1) $t_1 = t_2 > t_3$ (2) $t_1 < t_2 < t_3$
(3) $t_1 > t_2 > t_3$ (4) $t_1 = t_2 = t_3$

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19. A point mass is subjected to two simultaneous displacement in x-direction, $x_1(t) = A\sin(\omega t)$ and $x_2(t) = A\sin\left(\omega t + \frac{2\pi}{3}\right)$. Adding a third sinusoidal displacement $x_3(t) = B\sin(\omega t + \phi)$ brings mass to complete rest, then values of B and ϕ are

(1) $\sqrt{2}A, \frac{3\pi}{4}$ (2) $A, \frac{4\pi}{3}$

(3) $\sqrt{3}A, \frac{5\pi}{6}$ (4) $A, \frac{\pi}{3}$

20. The displacement of a harmonic oscillator is given by equation, $x = a\sin(\omega t) + b\cos(\omega t)$, then the amplitude of oscillator is (symbols have their usual meaning)

(1) a (2) b

(3) $\sqrt{a^2 + b^2}$ (4) $\sqrt{a^2 - b^2}$

21. Which function among the given options represents a non-periodic motion?

(1) $\sin(\omega t) + \cos(\omega t)$ (2) $\log_e(\omega t)$

(3) $e^{-\omega t}$ (4) Both (2) and (3)

22. A transverse wave is travelling along a stretched string. If tension in the string is made three times, then ratio of initial to final speed of transverse wave along stretched string, will be

(1) $1:\sqrt{3}$ (2) 1 : 2

(3) 1 : 1 (4) $\sqrt{3}:1$

23. The beat frequency, when two sound waves of frequency 450 Hz and 446 Hz are superposed will be

(1) 4 Hz (2) 5 Hz

(3) 2 Hz (4) 7 Hz

24. The average velocity of a particle executing SHM in one complete oscillation is

(1) $\frac{A\omega}{2}$ (2) $A\omega$

(3) $\frac{A\omega^2}{2}$ (4) Zero

25. On centigrade scale the temperature of a body increases by 45 degrees, then the increase in temperature on Fahrenheit scale will be

(1) 50° (2) 81°

(3) 45° (4) 30°

26. The power radiated by a black body is P and wavelength corresponding to maximum intensity radiation is λ_0. If the temperature of the body is now changed so that its wavelength corresponding to maximum intensity radiation becomes $\frac{\lambda_0}{2}$, then power radiated by it becomes nP. The value of n will be

(1) 81 (2) 256

(3) 8 (4) 16

27. A sinusoidal wave is given by, $y = A\sin(kx - \omega t)$. The ratio of maximum particle velocity to wave velocity will be (All symbols have their usual meaning)

(1) 1 (2) ω

(3) $A\omega$ (4) kA

28. If the speed of sound in hydrogen gas at NTP is 1328 m/s, then the speed of sound (at NTP) in a mixture of hydrogen and oxygen gases in the ratio 4 : 1 by volume will be

(1) 664 m/s (2) 158 m/s

(3) 365 m/s (4) 1270 m/s

29. The difference between the frequencies of the third and seventh harmonic of a closed organ pipe is 200 Hz, then its fundamental frequency will be

(1) 100 Hz (2) 50 Hz

(3) 25 Hz (4) 12.5 Hz

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30. Two sinusoidal waves having equations $y_1 = A\sin\left(\omega t - Kx - \frac{\pi}{6}\right)$ and $y_2 = A\sin\left(\omega t - Kx + \frac{\pi}{6}\right)$ are superimposed, then the equation of their resultant wave is

(1) $\frac{A}{\sqrt{3}}\sin(\omega t - Kx)$

(2) $A\sqrt{3}\sin(\omega t - Kx)$

(3) $A\sqrt{3}\sin\left(\omega t - Kx + \frac{\pi}{2}\right)$

(4) $\frac{A}{\sqrt{3}}\sin\left(\omega t - Kx - \frac{\pi}{2}\right)$

31. Consider following statement out of which one is labelled as **Assertion (A)** and other as **Reason (R)**.

Assertion (A): Among solids, liquids and gases speed of sound is maximum in solids.

Reason (R): Solids have very small coefficient of elasticity.

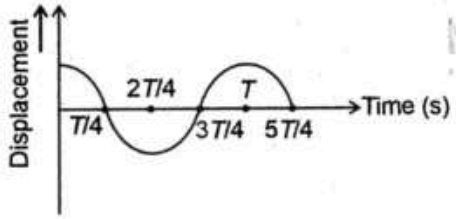
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

32. The displacement-time graph of a particle executing SHM is given in figure.



Based on above information consider the following statements.

a. The net force on the particle is zero at $t = \frac{3T}{4}$.

b. The magnitude of acceleration of the particle is maximum at $t = T$.

c. The speed of particle is maximum at $\frac{T}{4}$.

d. The potential energy is equal to kinetic energy of oscillation at $t = T$

Choose the correct option based on above information.

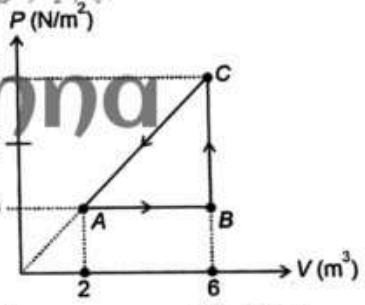
(1) Both statements a and d are correct

(2) Both statements b and d are correct

(3) Statements a, b and c are correct

(4) All statements are correct

33. An ideal gas follows a thermodynamic cyclic process as shown in figure. The internal energy of gas at point A is 10 J while that at B it is 30 J. If heat absorbed by gas during process BC is 120 J, then heat rejected in process CA will be



(1) 120 J (2) 300 J

(3) 100 J (4) 200 J

34. The change in internal energy of a gaseous system, if heat supplied to the gaseous system is 150 J and work done by the system is 80 J will be

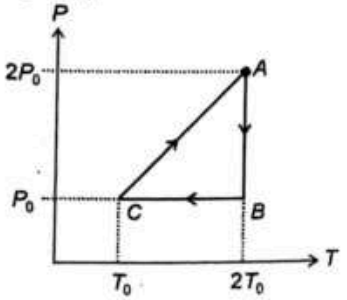
(1) 70 J (2) 80 J

(3) 100 J (4) Zero

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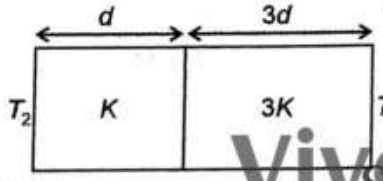
35. Three moles of an ideal gas undergoes a cyclic process as shown in figure. The work done by the gas during the process AB will be



(1) $3RT_0\ln(2)$ (2) $5RT_0\ln(1/2)$

(3) $6RT_0\ln(2)$ (4) $2RT_0\ln(2)$

36. Two materials having coefficient of thermal conductivity as 'K' and '$3K$' and thickness 'd' and '$3d$' respectively, are joined to form a slab as shown in figure. The temperature of the outer surfaces are 'T_2' and 'T_1', ($T_2 > T_1$), then temperature at interface is



(1) $\frac{T_1+T_2}{2}$ (2) $\frac{T_1+3T_2}{5}$

(3) $\frac{T_2+3T_1}{5}$ (4) $\frac{T_1+2T_2}{3}$

37. The specific heat of a substance at temperature t is given as, $\delta = at^2 + bt + c$, then the amount of heat required to raise temperature of a unit mass of a substance from zero to t_0 will be

(1) $at_0^3 + bt_0^2 + ct_0$ (2) $\frac{at_0^3}{3} + \frac{bt_0^2}{2} + ct_0$

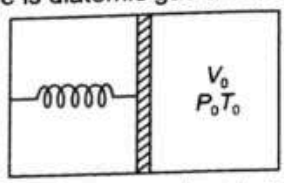
(3) $\frac{at_0^3}{3} + \frac{bt_0^2}{2} + \frac{ct_0}{2}$ (4) $\frac{at_0^3 + bt_0^2 + ct_0}{2}$

38. In a certain process, x amount of steam at 100°C becomes water at 100°C which converts y amount of ice at 0°C into water at 100°C, then the ratio x/y will be

(1) $\frac{2}{3}$ (2) 3

(3) $\frac{1}{3}$ (4) 1

39. A container has tight fitting movable piston which can slide without friction. The compartment containing spring has vacuum, and to right of the piston there is diatomic gas as shown in the figure.



Gas in the right compartment has pressure P_0, volume V_0 and temperature T_0. Natural length of the spring is equal to the length of the container. Find heat capacity of the system for the given state. (Neglect heat capacities of material of spring, piston and container).

(1) $\frac{P_0V_0}{T_0}$ (2) $\frac{2P_0V_0}{T_0}$

(3) $\frac{3P_0V_0}{T_0}$ (4) $\frac{4P_0V_0}{T_0}$

40. A point source of heat is placed at centre of spherical shell of inner radius a and outer radius b. The thermal conductivity of material of shell is K. If the temperature of inner and outer surface at steady state are T_1 and T_2 respectively ($T_1 > T_2$), then power of point source of heat will be

(1) $\frac{4\pi Kab(T_1-T_2)}{(b-a)}$ (2) $\frac{4\pi Kab(T_1-T_2)}{(b+a)}$

(3) $\frac{4\pi Kab(T_1+T_2)}{(b-a)}$ (4) $\frac{4\pi ab(T_1-T_2)}{K(a+b)}$

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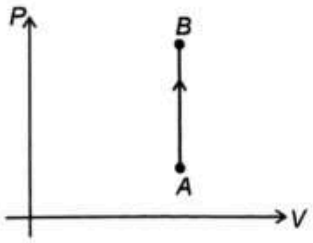
41. An ideal monoatomic gas is taken through a process in which $dQ = 4dU$, then molar heat capacity of the gas for the process is

(Where dQ represents heat interaction and dU represents change in internal energy)

(1) R (2) $2R$

(3) $6R$ (4) $4R$

42. Pressure (P) – volume (V) graph of a thermodynamic gaseous process moving from A to B is shown in the figure given below.



Based on above information, the correct option is (Symbols have their usual meaning)

(1) During process $A - B$, $\Delta P = 0$

(2) During process $A - B$, $\Delta Q = 0$

(3) During process $A - B$, $\Delta U = 0$

(4) During process $A - B$, $W = 0$

43. Adiabatic compression of an ideal gas causes

(1) The temperature to decrease

(2) The temperature to increase

(3) No change in temperature occurs

(4) Zero work done

44. If α moles of an ideal gas undergoes adiabatic process and changes its state from (P_1, V_1) to (P_2, V_2), then change in internal energy of gas will be (Assume the adiabatic index to be γ)

(1) $\frac{P_2V_2 - P_1V_1}{\gamma - 1}$

(2) $\frac{P_1V_1 - P_2V_2}{\gamma - 1}$

(3) $\frac{P_2V_2 + P_1V_1}{\gamma - 1}$

(4) $\frac{P_1V_2 - P_2V_1}{\gamma - 1}$

45. The equation of longitudinal stationary wave in third harmonic for a closed organ pipe is $y = 3(\sin\pi x)(\cos\pi t)$



Here x is in meter and t is in second. Based upon the above information match the Column I with Column II.

	Column I		Column II
(A)	Length of pipe	(P)	1.5 m
(B)	Wavelength	(Q)	1 m
(C)	Separation between consecutive nodes	(R)	2 m

(1) A-P; B-R; C-Q

(2) A-R; B-P; C-Q

(3) A-R; B-Q; C-P

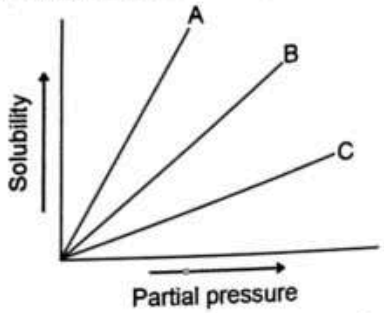
(4) A-P; B-Q; C-R

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[CHEMISTRY]

46. Solubilities of 3 gases A, B and C are plotted against partial pressure of the respective gas at constant temperature. The correct order of K_H for these gases is (K_H = Henry's law constant)



(1) A > B > C (2) C > B > A

(3) B > C > A (4) C > A > B

47. Given below are two statements.

Statement (I): A catalyst can change the rate of reaction of both spontaneous and non-spontaneous reaction.

Statement (II): For a chemical reaction with rise in temperature by 10°C, the rate constant is nearly doubled.

In the light of above statements, choose the **correct** answer from the options given below.

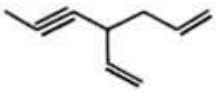
(1) Both statement I and statement II are true

(2) Statement I is true but statement II is false

(3) Statement I is false but statement II is true

(4) Both statement I and statement II are false

48. The correct IUPAC name of the below given molecule is



(1) 3-Propynylhexa-1,5-diene

(2) 4-Ethenylhept-6-en-2-yne

(3) 3-Propynylhepta-1,5-diene

(4) 4-Ethenylhept-1-en-5-yne

49. Match List-I with List-II.

	List-I (Aqueous solution and degree of dissociation)		**List-II (Value of van't Hoff factor)**
a.	$Al_2(SO_4)_3$, α = 60%	(i)	1.9
b.	NaCl, $\alpha \approx$ 90%	(ii)	3.1
c.	Na_3PO_4, α = 70%	(iii)	1.5
d.	$MgSO_4$, α = 50%	(iv)	3.4

Choose the **correct** match.

(1) a-(iv), b-(i), c-(ii), d-(iii)

(2) a-(i), b-(ii), c-(iii), d-(iv)

(3) a-(iv), b-(iii), c-(ii), d-(i)

(4) a-(ii), b-(iii), c-(i), d-(iv)

50. For the reaction,

$$2H_2 + 2NO \longrightarrow N_2 + 2H_2O$$

The following mechanism has been suggested:

(a) $2NO \rightleftharpoons N_2O_2$

(b) $N_2O_2 + H_2 \xrightarrow{k} N_2O + H_2O$ (slow)

(c) $N_2O + H_2 \longrightarrow N_2 + H_2O$

The overall order of given reaction is

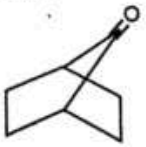
(1) –1 (2) 3

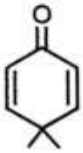
(3) 2 (4) Zero

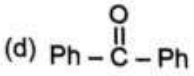
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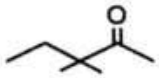
51. Consider the following compounds

(a) 

(b) 

(c) 

(d) 

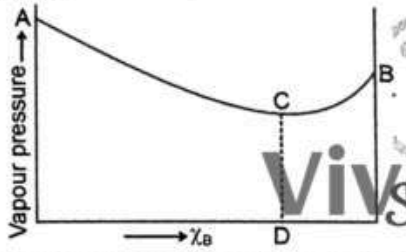
(e) 

The set of molecules which exhibit tautomerism is

(1) (a), (b) and (e) only (2) (b), (c) and (d) only

(3) (c) and (d) only (4) (a) and (e) only

52. The diagram given below is a vapour pressure composition diagram for a binary solution of A and B. Based on the diagram, consider the given statements.



(a) Binary mixture of A and B can form maximum boiling azeotropic at a specific composition.

(b) The intermolecular attractive forces between A-A and B-B are stronger than those between A-B.

(c) $\Delta_{mix}H < 0$ and $\Delta_{mix}V < 0$

(d) A and B can be n-hexane and n-heptane respectively.

(e) A and B can be ethanol and acetone respectively.

The set containing **incorrect** statements about binary mixture is

(1) (b) and (c) only (2) (a) and (c) only

(3) (b), (d) and (e) only (4) (a) and (e) only

53. In Carius method of estimation of halogen, 0.8 g of an organic compound gave 0.846 g of AgBr. The percentage of bromine in the compound is (Molar mass of AgBr = 188 g mol^{-1})

(1) 27% (2) 45%

(3) 52% (4) 35%

54. Consider the given statements.

(a) Nitrogen and phosphorus present in an organic compound can be detected by Lassaigne's test.

(b) AgBr is sparingly soluble in ammonium hydroxide solution.

(c) Lead sulphide is a yellow colour compound.

The correct statements are

(1) (b) and (c) only (2) (a), (b) and (c)

(3) (a) and (c) only (4) (a) and (b) only

55. The pH of 1 M solution of a weak monobasic acid (HA) is 1. The value of van't Hoff factor for this solution is

(1) 2.5 (2) 1.01

(3) 1.1 (4) 2.01

56. For the reaction $2NO + Cl_2 \longrightarrow 2NOCl$ at 300 K, the following data were obtained.

Experiment number	Initial concentration		Initial Rate
	[NO]	$[Cl_2]$	
1.	0.01	0.01	1.2×10^{-4}
2.	0.01	0.02	2.4×10^{-4}
3.	0.02	0.02	9.6×10^{-4}

Based on the given information the rate constant for the reaction is

(1) 120 s^{-1} (2) 100 $mol^{-2}\ L^2\ s^{-1}$

(3) 120 $mol^{-2}\ L^2\ s^{-1}$ (4) 70 $mol^{-2}\ L^2\ s^{-1}$

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57. The total number of structural isomers of $C_4H_{10}O$ is
(1) 6 (2) 5
(3) 8 (4) 7

58. For a catalysed reaction activation energy is lowered by 24.492 kJ/mol. The ratio of the rate of the catalysed reaction to that of the uncatalyzed reaction at 750 K is ($e^4 = 54.6$)
(1) 27.3 (2) 54.6
(3) 7.38 (4) 108.12

59. N_2O_5 decomposes to NO_2 and O_2 and follows 1st order kinetics.

$$N_2O_5(g) \longrightarrow 2NO_2(g) + \frac{1}{2}O_2(g)$$

Time	Total pressure
0	40 mm Hg
50 min	70 mm Hg
100 min	P_T

The value of P_T is
(1) 85 mm Hg (2) 75 mm Hg
(3) 100 mm Hg (4) 57 mm Hg

60. Match List-I with List-II

	List-I (Species)		List-II (Colour)
a.	$Fe_4[Fe(CN)_6]_3 \cdot xH_2O$	(i)	Blood red
b.	$[Fe(CN)_5NOS]^{4-}$	(ii)	Violet
c.	$[Fe(SCN)]^{2+}$	(iii)	Yellow
d.	$(NH_4)_3PO_4 \cdot 12MoO_3$	(iv)	Prussian blue

Choose the correct match
(1) a-(iv), b-(ii), c-(i), d-(iii)
(2) a-(i), b-(ii), c-(iii), d-(iv)
(3) a-(iv), b-(iii), c-(ii), d-(i)
(4) a-(ii), b-(i), c-(iv), d-(iii)

61. Given below are two statements.

Statement (I): Sodium amalgam is a homogenous mixture of sodium and mercury where solvent is solid and solute is liquid.

Statement (II): Molality and mole fraction are temperature dependent quantity.

In the light of above statements, choose the **correct** answer from options given below.
(1) Statement (I) is true but statement (II) is false
(2) Statement (I) is false but statement (II) is true
(3) Both statement (I) and statement (II) are true
(4) Both statement (I) and statement (II) are false

62. If 90% of first order reaction completes in 200 min, then the half-life of reaction is approximately
(1) 2 hour (2) 2.5 hour
(3) 1 hour (4) 1.5 hour

63. Match List-I with List-II.

	List-I (Species)		List-II (Characteristic)
a.	Tropone	(i)	Heterocyclic non-aromatic
b.	Tetrahydrofuran	(ii)	Heterocyclic aromatic
c.	Thiophene	(iii)	Benzenoid aromatic
d.	Anthracene	(iv)	Non-benzenoid aromatic

Choose the **correct** match.
(1) a-(i), b-(ii), c-(iii), d-(iv)
(2) a-(iv), b-(iii), c-(ii), d-(i)
(3) a-(ii), b-(i), c-(iii), d-(iv)
(4) a-(iv), b-(i), c-(ii), d-(iii)

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64. A solution is 9.8% H_2SO_4 by mass. The molality of this solution will be

(1) 1.1 m (2) 2.2 m

(3) 3.3 m (4) 0.1 m

65. The activation energy for a certain reaction is 40 kcal/mol at 27°C. The fraction of molecules of reactants having energies equal to or greater than activation energy is

(1) $e^{+16.03}$ (2) $e^{-0.06}$

(3) $e^{-66.67}$ (4) $e^{-16.03}$

66. A sample of 0.2 g of an organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed in 20 mL of 0.1 M H_2SO_4. The residual acid required 10 mL of 0.1 M NaOH solution for neutralisation. The percentage of nitrogen in the compound is

(1) 21% (2) 7%

(3) 28% (4) 42%

67. Two liquids A and B have vapour pressure in the ratio $P_A^\circ : P_B^\circ = 1:3$ at a certain temperature. The mole fraction of A and B in their ideal solution is in the ratio of 2 : 7 respectively. The mole fraction of A in vapour phase is

(1) $\frac{21}{23}$ (2) $\frac{2}{21}$

(3) $\frac{2}{23}$ (4) $\frac{6}{13}$

68. Consider the following statements

(a) Inversion of cane sugar in an acidic medium is a pseudo first order reaction.

(b) A catalyst never participates in a chemical reaction.

(c) For complex reaction, order is given by the slowest step of the reaction.

(d) Molecularity of a reaction can be zero or a non integer.

(e) Decomposition of N_2O is an example of zero order reaction.

The incorrect statements are

(1) (b), (d) and (e) only

(2) (d) and (e) only

(3) (a) and (c) only

(4) (a), (b) and (c) only

69. Given below are two statements. One is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A): The decomposition of gaseous ammonia on a hot Pt surface is a zero order reaction at high pressure.

Reason (R): At high pressure, the metal surface gets saturated with gas molecules.

In the light of above statements, choose the **correct** answer from options given below.

(1) (A) is true but (R) is false

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) (A) is false but (R) is true

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

70. In sulphur estimation, 1.25 g of an organic compound gave 1.398 g of barium sulphate. The percentage of sulphur in the compound is (Molar mass of $BaSO_4$ = 233 g mol^{-1})

(1) 32.25% (2) 45%

(3) 48% (4) 15.36%

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71. For which of the following solution(s), abnormal molar mass of the solute is observed?

(a) Aqueous solution of CH_3COOH

(b) Aqueous solution of urea

(c) Aqueous solution of glucose

(d) CH_3COOH in benzene

(e) Aqueous solution of K_2SO_4

Choose the correct option.

(1) (b) and (c) only (2) (a), (b) and (e) only

(3) (a), (d) and (e) only (4) (e) only

72. If a 1st order reaction follows the given equation $\log k = \frac{-3000}{T} + 5$ where k is rate constant and T is temperature, then the value of pre-exponential factor 'A' and activation energy respectively are

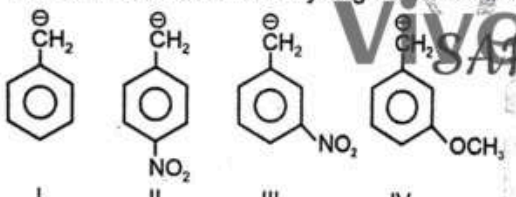
(1) 10^5 and 57.44 kJ/mol

(2) 10^6 and 41.3 kJ/mol

(3) e^5 and 10 kJ/mol

(4) e^6 and 5 kJ/mol

73. The correct order of stability of given carbanions is



I II III IV

(1) III > IV > I > II (2) II > IV > III > I

(3) I > II > III > IV (4) II > III > IV > I

74. Consider the following statements.

(a) Mixture of chloroform and acetone forms a solution which shows positive deviation from Raoult's law.

(b) An ideal solution cannot form azeotropic mixture.

(c) Reverse osmosis is always a spontaneous process.

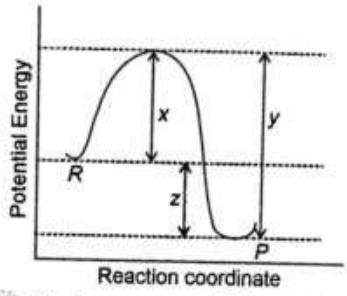
(d) On addition of small amount of HgI_2 in a solution of KI, boiling point of resultant solution increases.

Correct statement(s) is/are

(1) (a) and (d) only (2) (b) and (d) only

(3) (a), (c) and (d) only (4) (b) only

75. Consider the following statements for mentioned graph



(a) x = Activation energy of forward reaction

(b) y = Activation energy of backward reaction

(c) z = Enthalpy of reaction

(d) Given reaction is endothermic

(e) x + z = Threshold energy

The **incorrect** statements are

(1) (a), (b) and (c) only

(2) (d) and (e) only

(3) (b) and (e) only

(4) (a), (c) and (d) only

76. The method which is used to purify liquids with very high boiling points and those, which decompose at or below their boiling points is

(1) Steam distillation

(2) Distillation under reduced pressure

(3) Fractional distillation

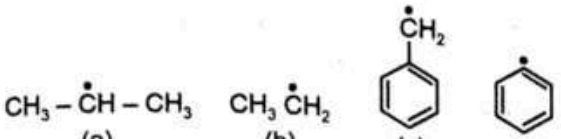
(4) Differential extraction

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77. If blood cells are kept in a solution having 1.4% NaCl (mass/volume), then the **correct** statement is
(1) Water will flow out of the cells and they would shrink
(2) Water will flow into the cells and they would swell
(3) The solution is hypotonic to cell fluid
(4) No osmosis would occur as the solution is isotonic with cell fluid

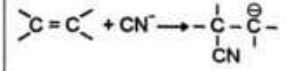
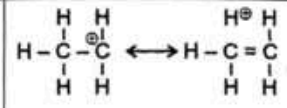
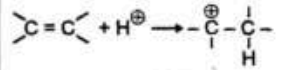
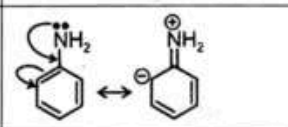
78. Consider the following species.



(a) (b) (c) (d)

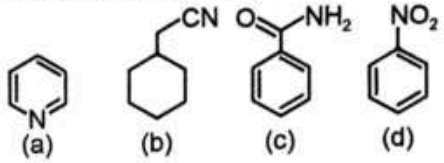
The correct order of stability of the given radicals is
(1) (c) > (d) > (a) > (b) (2) (a) > (c) > (d) > (b)
(3) (c) > (a) > (b) > (d) (4) (d) > (c) > (a) > (b)

79. Match List-I with List-II.

	List-I (Displacement of electron)		List-II (Electronic effect)
a.		(i)	+E
b.		(ii)	–E
c.		(iii)	+R
d.		(iv)	No bond resonance.

Choose the correct match
(1) a-(ii), b-(iv), c-(i), d-(iii)
(2) a-(i), b-(ii), c-(iii), d-(iv)
(3) a-(iv), b-(iii), c-(ii), d-(i)
(4) a-(ii), b-(iii), c-(iv), d-(i)

80. Consider the following compounds



(a) (b) (c) (d)

Kjeldahl method is applicable to
(1) (a) and (b) only (2) (b) and (d) only
(3) (b), (c) and (d) only (4) (b) and (c) only

81. For a zero order reaction, 30% of reaction completes in 20 s. The half-life of reaction is
(1) 33.33 s (2) 14.3 s
(3) 20.3 s (4) 37.1 s

82. The correct order of stability of following resonating structures is
(I) $CH_2 = CH - CH = CH_2$
(II) $\overset{\oplus}{C}H_2 - CH = CH - \overset{\ominus}{C}H_2$
(III) $\overset{\oplus}{C}H_2 - CH = CH - \overset{\ominus}{C}H_2$
(1) I > II > III (2) III > I > II
(3) I > III > II (4) II > I > III

83. How many grams of concentrated nitric acid solution should be used to prepare 800 mL of 3.0 M HNO_3?
(Given concentrated nitric acid is 70% $\left(\frac{w}{w}\right)$ HNO_3)
(1) 181 g (2) 216 g
(3) 436 g (4) 321 g

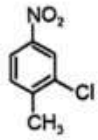
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84. Consider the following statements.

(a) Hyperconjugation effect is observed in allyl $\left(CH_2 = CH - \overset{\oplus}{C}H_2\right)$ carbocation.

(b) Lone pair of electron on N in pyridine is not delocalised.

(c) The IUPAC name of  is 2-Chloro-1-methyl-4-nitrobenzene.

The **correct** statement(s) is/are

(1) (a), (b) and (c) (2) (b) and (c) only

(3) (a) and (b) only (4) (c) only

85. Among the given mixture, the one with highest boiling point is

(1) Na_2SO_4, 0.1 m, 20% dissociation

(2) CH_3COOH, 0.2 m, 10% dissociation

(3) K_3PO_4, 0.3 m, 10% dissociation

(4) $MgCl_2$, 0.1 m, 30% dissociation

86. For a first order reaction, rate of reaction at 30 min from start is 0.03 M s^{-1} and at 60 min from start is 0.02 M s^{-1}, then the half-life of reaction will be nearly

(1) 51 min (2) 30 min

(3) 45 min (4) 27 min

87. Consider the following statements

(a) Column chromatography is a type of adsorption chromatography.

(b) Alumina can be used as an adsorbent in adsorption chromatography.

(c) Different fractions of crude oil in petroleum industry can be separated by fractional distillation.

The correct statements are

(1) (a), (b) and (c) (2) (a) and (b) only

(3) (a) and (c) only (4) (b) and (c) only

88. Given below are two statements.

Statement (I): The value of Cryoscopic constant depends upon the nature of solvent.

Statement (II): Cryoscopic constant (K_f) is given by $\frac{R \times M_1 \times T_f^2}{10^3 \times \Delta_{fus}H}$, where R is gas constant, T_f = freezing point of pure solvent, M_1 = molar mass of solvent and $\Delta_{fus}H$ = Enthalpy of fusion.

In the light of above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are true

(2) Statement I is true but statement II is false

(3) Statement I is false but statement II is true

(4) Both statement I and statement II are false

89. For a chemical reaction A + B → Product, if the concentration of A is doubled, the rate of reaction becomes half and if both the concentration of A and B are doubled, the rate of reaction becomes doubled. The overall order of reaction will be

(1) 2 (2) 1

(3) 1.5 (4) Zero

90. The set of species that can act as electrophile is

(1) BF_3, $\overset{+}{N}O_2$ and NH_3

(2) $CH_3 - \overset{+}{C} = O$, $CH_3 - \overset{+}{C}H_2$ and $AlCl_3$

(3) Cl^+, SO_3 and OH^-

(4) $\overset{+}{C}H_3$, $(CH_3)_2NH$ and $\overset{+}{B}r$

Space for Rough Work

[BOTANY]

91. During the conversion of which of the following substrates a molecule of GTP is synthesized in TCA cycle?
(1) Succinyl CoA → Succinic acid
(2) Citric acid → Isocitric acid
(3) Oxalosuccinic acid → α-ketoglutaric acid
(4) Pyruvate → Acetyl CoA

92. Select the **odd** one w.r.t. tomato.
(1) Growth in carbon dioxide enriched atmosphere leads to higher yields.
(2) Parthenocarpy is induced by auxins
(3) Fruit ripening is hastened by the usage of ethephon
(4) They have RuBisCO in bundle sheath cells

93. Which of the following reactions is catalysed by an enzyme that is utilized in both mitochondrial ETS and Tricarboxylic acid cycle?
(1) Succinate → Fumarate
(2) α-ketoglutaric acid → Succinyl CoA
(3) 3-Phosphoglycerate → 2-phosphoglycerate
(4) Malic acid → Oxaloacetic acid

94. Non-cyclic photophosphorylation differs from cyclic photophosphorylation, as the former
(1) Yields only ATP
(2) Requires the photosystem having reaction centre P700
(3) Requires external electron donor
(4) Takes place in stroma lamellae

95. Which of the following expression is correct w.r.t. RQ of different respiratory substrates?
(1) Tripalmitin > proteins
(2) Proteins = fats
(3) Organic acids > glucose
(4) Glucose < tripalmitin

96. To stimulate flowering in pineapples, which set of PGRs can be used?
(1) Ethylene and Auxin
(2) GA and ABA
(3) Zeatin and dormin
(4) Inhibitor-B and abscission II

97. Select the **mismatched** pair
(1) Cytoplasm – Site of the pathway that is the only process of respiration in anaerobic organisms
(2) Mitochondrial matrix – Site of the reaction where condensation of acetyl group with oxaloacetic acid (OAA) and water occurs to yield a 5C compound
(3) Mitochondrial inner membrane – Site of placement of cytochrome *c* oxidase
(4) Cytosol – Site of partial oxidation of glucose

98. Select the **correct** statement w.r.t. xanthophylls.
(1) It appears bright or blue green in the chromatogram.
(2) It can act as an accessory pigment.
(3) It shows maximum absorption between 500-600 nm
(4) Its photooxidation is prevented by chlorophyll a.

99. Both auxin and ethylene can
(1) Suppress germination of peanut seeds
(2) Induce horizontal growth of seedlings
(3) Stimulate root growth
(4) Prevent senescence and abscission of plant organs especially of leaves

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100. Match column-I with column-II and select the correct option.

	Column -I		Column-II
I.	Phosphoglycolate	a.	Undergoes decarboxylation in bundle sheath cells
II.	3-phosphoglycerate	b.	Formed when most abundant enzyme in the world interacts with oxygen
III.	Aspartic acid	c.	Regenerated in the mesophyll cell by using ATP
IV.	Ribulose-1, 5 bisphosphate	d.	Product of the most crucial step of the Calvin cycle

	I	II	III	IV
(1)	a	d	c	b
(2)	b	a	d	c
(3)	c	b	a	d
(4)	b	d	a	c

101. All of the following delay senescence, **except**

(1) Gibberellin (2) Cytokinin

(3) Auxin (4) ABA

102. Which of the following is the high energy chemical intermediate(s) formed during light reaction of photosynthesis?

(1) ADP and NADH (2) NADPH

(3) O_2 (4) $FADH_2$

103. The PGR that was discovered while studying foolish seedling disease of rice, is also known for

(1) Inducing internode elongation just prior to flowering in cabbages

(2) Hastening the maturity period in conifers to avoid early seed production

(3) Suppressing of seed germination

(4) Decreasing in the stem length of sugarcane

104. Maize shows

(1) Carboxylation only in mesophyll cells

(2) Greater productivity of biomass due to formation of phosphoglycolate by activity of RuBisCO.

(3) CO_2 saturation only beyond 450 $\mu l L^{-1}$.

(4) Wreath type of arrangement of bundle sheath cells.

105. Read the following Assertion (A) and Reason (R) statements and select the **correct** option.

Assertion (A): Plants retain the capacity for unlimited growth throughout their life.

Reason (R): Plants show the presence of meristems at certain locations in their body.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

106. Select the **odd** one w.r.t. alcoholic fermentation.

(1) Evolution of CO_2

(2) Involvement of pyruvic acid decarboxylase

(3) Takes place in cytoplasm

(4) Gives net yield of 2 NADH for oxidation of each molecule of glucose

SAF_LYFFF
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107. Which of the following scientists provided evidence for production of glucose when plants grow?
(1) Joseph Priestley (2) Jan Ingenhousz
(3) Julius von Sachs (4) T.W Engelmann

108. Read the following statements and select the correct option.
Statement A: Efficiency index is the measure of the ability of the plant to produce new plant material.
Statement B: Macro and micro-essential elements are required by plants for the synthesis of protoplasm and act as source of energy.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

109. Read the following statements:
a. The growth of a pollen tube is measured in terms of its length.
b. Cells positioned away from root apical meristems differentiate as epidermis, while those pushed to the periphery mature as root-cap cells.
c. Measurement and the comparison of total growth per unit time is called the absolute growth rate.
d. Charles Darwin and his son Francis Darwin observed that the coleoptiles of oats responded to unilateral illumination by growing towards the light source.
e. Turgidity of cells helps in extension growth.
Select the **correct** ones:
(1) a, c and e only
(2) b and d only
(3) a, b and c only
(4) c, d and e only

110. Which of the following PGR stimulates the closure of stomata and increases the tolerance of plants to various kinds of stresses?
(1) Ethylene (2) Cytokinin
(3) GA (4) ABA

111. All of the following are the plant factors affecting photosynthesis, **except**
(1) Orientation of leaves
(2) Internal CO_2 concentration
(3) Atmospheric temperature
(4) Amount of chlorophyll

112. Apical hook formation in dicot seedlings is induced by
(1) Auxin (2) Cytokinin
(3) Gibberellins (4) Ethylene

113. During light reaction, ATP synthesis in the chloroplasts is a direct outcome of
(1) Formation of proton gradient across the thylakoid membrane
(2) Removal of protons from the stroma while transporting electrons
(3) The breakdown of the proton gradient causing a conformational change in the CF_1 particle
(4) The release of oxygen and production of high energy intermediates

114. Which of the following is the key product of glycolysis?
(1) Pyruvate (2) Ethanol
(3) Lactic acid (4) Fructose

115. The PGR that was isolated from the autoclaved herring sperm DNA is
(1) Auxin (2) Gibberellin
(3) Kinetin (4) Ethylene

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116. These are the following enzymes that participate during the oxidation of disaccharide composed of glucose and fructose via respiration

a. Enolase
b. Phosphofructokinase
c. Invertase
d. Hexokinase
e. Aldolase

Select the option that depicts the **correct** chronological sequence of participation of these enzymes during aerobic respiration pathway.

(1) a, c, e, b, d (2) c, d, b, a, e
(3) c, d, b, e, a (4) c, b, a, e, d

117. Removal of shoot tips (decapitation) suppresses the effect of

(1) Auxin (2) Cytokinin
(3) GA (4) ABA

118. In a photosystem

(1) LHC are made up of hundreds of pigment molecules only.
(2) Various pigments help to make photosynthesis more efficient by absorbing wavelengths of light in green region.
(3) All the pigments (except one molecule of chlorophyll a) forms antennae.
(4) Reaction centre is made up of chlorophyll c having absorption peak at 700 nm.

119. __(A)__ et al., identified and crystallised the cytokinesis promoting active substance that they termed kinetin. Select the **correct** option to fill (A).

(1) Miller (2) F. W. Went
(3) H.H. Cousins (4) E. Kurosawa

120. Match column-I and column-II and select the **correct** option.

	Column -I		Column -II
I.	Auxin	a.	Promotes rapid internode/petiole elongation in deep water rice plants
II.	Cytokinin	b.	Controls xylem differentiation and helps in cell division
III.	Ethylene	c.	Causes fruits like apple to elongate and improves its shape.
IV.	Gibberellin	d.	Promotes nutrient mobilization which helps in the delay of leaf senescence

	I	II	III	IV
(1)	a	d	c	b
(2)	b	a	d	c
(3)	c	b	a	d
(4)	b	d	a	c

121. Which of the following PGRs possess terpenes in its chemical composition?

(1) C_2H_4
(2) Gibberellin
(3) N^6 -furfuryl amino purine
(4) Indole Acetic Acid

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122. Read the following Assertion (A) and Reason (R) statements and select the **correct** option.

Assertion (A): In C_4 plants, photorespiration does not occur.

Reason (R): C_4 plants have a mechanism that increases the concentration of CO_2 at the enzyme site (RuBisCO).

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

123. Glycerol would enter the respiratory pathway after being converted to

(1) Acetyl CoA

(2) PGAL

(3) Amino acids

(4) Pyruvate

124. Which of the following molecules act as a final hydrogen acceptor in mitochondrial ETS?

(1) Oxygen

(2) NADH

(3) $FADH_2$

(4) ATP

125. All of the following features are shown by the cells in the meristematic phase, **except**

(1) Rich in protoplasm

(2) Increased vacuolation

(3) Possess large conspicuous nuclei

(4) Cell walls are primary in nature

126. Bell pepper and sorghum can be differentiated from each other as

a. For the former, current availability of CO_2 levels are limiting.

b. The latter responds to high CO_2 conditions under low light conditions.

c. In the former, water stress leads to reduction in the CO_2 availability.

d. In the former, RuBisCO can show oxygenase activity.

e. In the former, pyruvate kinase is found in mesophyll cells.

The **correct** ones are:

(1) a, b and e only (2) a and d only

(3) b, c and d only (4) c and e only

127. Heterophyllous development due to environment can be seen in

(1) Cotton

(2) Coriander

(3) Buttercup

(4) Larkspur

128. Read the following statements and select the **correct** option.

Statement A: Light saturation occurs at 10 per cent of the full sunlight.

Statement B: Within the chloroplasts, the membranes are sites for the carbon reaction, while the photochemical pathway occurs in the stroma.

(1) Only statement A is correct.

(2) Only statement B is correct.

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

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129. Which of the following PGR is used to speed up the malting process in brewing industry?

(1) GA (2) ABA

(3) IAA (4) Zeatin

130. From the statements given below choose the **correct** ones:

A. Fermentation takes place in many prokaryotes, unicellular eukaryotes and in germinating seeds.

B. The initial stage of cellular respiration takes place in the cytoplasm.

C. Most cells of a plant have atleast a part of their surface in contact with air.

D. For each ATP produced, $2H^+$ passes through F_0 from the intermembrane space to the matrix against the electrochemical proton gradient.

E. The continued oxidation of acetyl CoA via the TCA cycle requires the continued replenishment of citric acid, the first member of the cycle.

(1) A, C and E only

(2) B, D and E only

(3) A, B and C only

(4) A, D and E only

131. All of the following can be used to overcome seed dormancy, **except**

(1) Chilling conditions

(2) Nitrates

(3) Scratching of seed coat of hard seeds

(4) Para-ascorbic acid

132. Select the **mismatched** pair.

(1) Mint plant – used as an experimental organism by Priestley to propose that plants restore to the air whatever breathing animals and burning candles removed.

(2) Aquatic plant - used as an experimental organism by Ingenhousz to propose that it is only the green part of the plants that could release oxygen.

(3) Brown alga - used as an experimental organism by Engelmann to propose the first action spectrum of photosynthesis

(4) Purple and green bacteria - used as an experimental organism by Cornelius van Niel to demonstrate that photosynthesis is essentially a light-dependent reaction in which hydrogen from a suitable oxidisable compound reduces carbon dioxide to carbohydrates.

133. Read the following steps of Hatch and Slack pathway:

a. Conversion of 3 carbon molecule to primary CO_2 acceptor

b. Conversion of first stable product into malic acid

c. Transport of 4-carbon organic acid into the cells encaging RuBisCO

d. Carboxylation reaction catalyzed by PEP carboxylase

e. Decarboxylation of C_4 acid

The **correct** sequential order of these events are:

(1) d → b → e → c → a

(2) a → b → c → d → e

(3) a → b → c → e → d

(4) d → b → c → e → a

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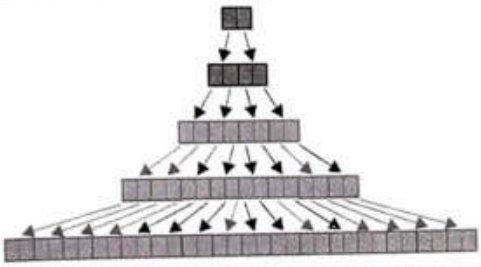
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134. How much additional molecules of NADH + H^+ will be produced if one molecule of Glyceraldehyde-3-phosphate is used as a respiratory substrate instead of one molecule of fumaric acid?

(1) 1 (2) 5

(3) 3 (4) 4

135. Observe the following figure:



Based on the above figure, certain statements have been made:

a. This type of growth results in a curve which is a characteristic of a living organism growing in a natural environment.

b. This type of growth is slowed down by limited nutrient supply.

c. This type of growth is exemplified by root elongating at a constant rate.

d. In this type of growth, following mitotic cell division, only one daughter cell continues to divide while the other differentiates and matures.

e. Mathematically, this growth can be expressed as $L_t = L_0 + rt$

How many of these statements is/are **correct**?

(1) Two (2) Three

(3) One (4) Five

[ZOOLOGY]

136. Animal possessing worm-like body with an excretory organ called proboscis gland belongs to the phylum

(1) Annelida

(2) Hemichordata

(3) Echinodermata

(4) Platyhelminthes

137. Choose the **incorrect** feature w.r.t. members of the phylum to which *Aplysia* belongs to.

(1) The anterior head region bears sensory tentacles

(2) They are unsegmented invertebrates

(3) Organ-system level of organisation is seen

(4) They are exclusively marine

138. Select the **incorrect** match from the following.

(1) Living fossil – *Limulus*

(2) Gregarious pest – *Locusta*

(3) Vector – *Apis*

(4) Economically important insect – *Bombyx*

139. Choose the option which is not true w.r.t. features of the members of a phylum whose females are generally longer than males and are pseudocoelomates.

(1) Possess muscular pharynx

(2) Have organ-system level of organisation

(3) Have excretory tube for removal of body waste

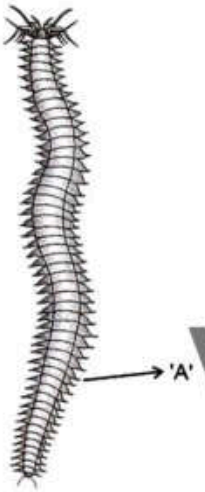
(4) Have well developed circulatory and respiratory system

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140. *Cucumaria* is similar to *Chaetopleura* with respect to
(a) Absence of larval stages during development
(b) Presence of a well-developed excretory system
(c) Being a dioecious non-chordate
(d) Having marine habitat
Choose the **correct** option.
(1) (a) and (c) only
(2) (b) and (d) only
(3) (c) and (d) only
(4) (a) and (d) only

141. Identify the given figure.



Choose the correct option w.r.t. the structure labelled as 'A'.
(1) The phylum to which the above given animal belongs to, is named on this structure
(2) Dorsal appendage that help in osmoregulation
(3) Lateral appendage that help in swimming
(4) Exclusively formed of longitudinal muscles

142. Consider the following statements:
Statement (A): Sponges are primitive unicellular animals having cellular level of organisation.
Statement (B): Choanocytes or collar cells line the spongocoel and the canals.
Select the correct option.
(1) Both statements (A) and (B) are correct
(2) Both statements (A) and (B) are incorrect
(3) Only statement (A) is incorrect
(4) Only statement (B) is incorrect

143. Read the following statements:
(a) Animals like annelids and arthropods, where the body can be divided into identical dorsal and ventral halves in only one plane, exhibit bilateral symmetry.
(b) Those animals in which the developing embryo exhibits mesoderm in between the ectoderm and endoderm, are called triploblastic animals.
(c) Closed type of blood circulation is seen in organisms belonging to the phylum Annelida and Chordata.
(d) The body of ctenophores and molluscs can be divided into two identical halves, in any plane that passes through the central axis of their body.
Choose the option representing the **true** statements only.
(1) (a) and (b) only
(2) (b) and (c) only
(3) (a) and (d) only
(4) (a), (b), (c) and (d)

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144. Choose the odd one w.r.t. fishes in which the pelvic fins of male members bear claspers.

(1) *Carcharodon* (2) *Torpedo*

(3) *Clarias* (4) *Trygon*

145. Which among the following is incorrect for pearl oyster?

(1) Economically beneficial

(2) Unsegmented organism

(3) Shell is reduced

(4) Marine habitat

146. Match the Column I with Column II.

	Column I		Column II
(a)	Limbless amphibian	(i)	*Hippocampus*
(b)	Oviparous mammal	(ii)	*Ornithorhynchus*
(c)	Aquatic arthropod	(iii)	*Palaemon*
(d)	Fish with pre-hensile tail	(iv)	*Ichthyophis*

Choose the **correct** option.

	(a)	(b)	(c)	(d)
(1)	(i)	(ii)	(iii)	(iv)
(2)	(iv)	(i)	(ii)	(iii)
(3)	(iv)	(ii)	(iii)	(i)
(4)	(i)	(iii)	(ii)	(iv)

147. Which of the following is/are **incorrect** match(es)?

(a) Metagenesis – *Obelia*

(b) Polyp form – *Adamsia*

(c) Medusa form – *Aurelia*

(d) Medusoid phase dominant – *Pennatula*

Choose the appropriate option.

(1) Only (a) (2) Only (d)

(3) (b) and (c) (4) (a) and (d)

148. Choose the **incorrect** statement.

(1) Unlike members of the class Chondrichthyes, placoid scales are absent in amphibians.

(2) *Ascidia* is a chordate that belongs to the super class Tetrapoda.

(3) Paired pectoral fins are present in cartilaginous fishes.

(4) The body of cyclostomes is devoid of paired fins.

149. Select the **incorrect** option w.r.t. mammals.

(1) They exhibit heterodont dentition.

(2) Their skin is unique in possessing hair.

(3) Pulmonary ventilation is seen in them.

(4) They are homeotherms due to presence of incomplete double circulation.

150. Tympanum represents ear in

(1) Salamander, Tree frog, Lamprey

(2) Wall lizard, Toad, Frog

(3) Vulture, Dolphin, Katla

(4) Wall lizard, Crocodile, Flying fox

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151. Read the given statements and select the **correct** option.

Statement (A): On land, throat-based ventilation and cutaneous respiration is characteristic of frogs.

Statement (B): All organisms of the class Mammalia possess hair and pinnae.

(1) Both statements (A) and (B) are correct
(2) Both statements (A) and (B) are incorrect
(3) Only statement (A) is correct
(4) Only statement (B) is correct

152. Choose the **correct** match from the following w.r.t. cockroaches.

(1) Tegmina – Prothoracic wings
(2) Labrum – Lower lip
(3) Forewings – Mesothoracic wings
(4) Maxillae – Possess grinding and incising regions

153. Which of the following sterna of a female cockroach are involved in the formation of its brood pouch?

(1) 6th, 7th and 9th
(2) 7th, 9th and 10th
(3) 7th, 8th and 9th
(4) 8th, 9th and 10th

154. Choose an **incorrect** statement w.r.t. the external morphology of *Periplaneta*.

(1) Head is triangular in shape and lies anteriorly at right angle to the longitudinal body axis.
(2) The entire body is covered by hard chitinous exoskeleton.
(3) In the head region, the brain is represented by supra-oesophageal ganglion which supplies nerves to all the mouth parts.
(4) A pair of thread-like antennae arise from membranous socket lying in front of the eyes.

155. How many of the following structures present in cockroaches are sensory in function?

Maxilla, Antenna, Jointed leg, Anal cerci

Choose the **correct** option.

(1) Three (2) One
(3) Four (4) Two

156. Read the following w.r.t. a ring of thin filamentous structures present at the junction of midgut and hindgut of *Periplaneta*.

(i) Ring consists of only 6-8 blind tubules
(ii) Secrete digestive juices
(iii) Help in removal of waste products directly from alimentary canal.
(iv) Each tubule of ring is lined by glandular and ciliated cells.

Choose the option that correctly represents the above facts as true (T) or false (F).

	(i)	(ii)	(iii)	(iv)
(1)	T	T	F	F
(2)	F	F	F	T
(3)	F	F	T	T
(4)	T	T	F	T

157. Which among the following is not **incorrect** regarding *Periplaneta*?

(1) A pair of spermatheca is present in the 6th abdominal segment of males.
(2) Sperms are transferred into female through spermatophores.
(3) Wing pads are present only in adult forms.
(4) In females, urecose glands help in excretion.

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158. The type of portal system which is common between frog and human is a special venous connection between
(1) Intestine and liver
(2) Liver and kidney
(3) Hindlimbs and kidney
(4) Kidney and heart

159. Consider the following statements w.r.t. adult frogs.
(a) They swim in water by powerful backward thrust of hindlimbs that ends with four digits.
(b) Branchial respiration is seen in them under water.
(c) Croaking sound is louder in male frogs due to the presence of vocal cords.
Select the option with **incorrect** statement(s) only.
(1) (a) and (b) only
(2) (a) only
(3) (a), (b) and (c)
(4) (b) and (c) only

160. Read the following
(a) Thyroid gland
(b) Parathyroid gland
(c) Pineal body
(d) Pancreatic islets
(e) Gonads
(f) Adrenal gland
(g) Pituitary gland
(h) Thymus gland
How many of the above are the prominent endocrine glands found in frogs?
(1) Five (2) Six
(3) Seven (4) Eight

161. Choose the **incorrect** statement w.r.t. nervous system of the frog.
(1) The nervous system is organised into a central and a peripheral nervous system
(2) There are 10 pairs of spinal nerves arising from the brain.
(3) Brain is enclosed in a bony structure called cranium.
(4) Hind brain consists of cerebellum and medulla oblongata but lacks pons.

162. **Assertion (A):** Larval stages of frogs have a relatively long digestive tract than their adult stages.
Reason (R): Tadpoles are herbivores and their plant based diet require a longer and greater surface area for digestion and absorption.
In the light of above statements, choose the **correct** option.
(1) (A) is true but (R) is false
(2) Both (A) and (R) are false
(3) Both (A) and (R) are true and (R) is the correct explanation of (A)
(4) Both (A) and (R) are true but (R) is not the correct explanation of (A).

163. Select the **incorrect** option w.r.t. *Rana tigrina*.
(1) Poikilotherm (2) Tetrapod
(3) Viviparous (4) Ureotelic

164. 'X' in male frog arise from testes, are 10-12 in number and enter the kidneys on their side and open into 'Y', which communicates with a tract that carries both sperm and urine.
Identify 'X' and 'Y' and choose the **correct** option.

	X	Y
(1)	Vasa diferentia	Urinogenital duct
(2)	Vasa diferentia	Bidder's canal
(3)	Vasa efferentia	Bidder's canal
(4)	Vasa diferentia	Cloaca

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165. The frogs are similar to the humans due to the presence of
(1) Nucleated RBCs
(2) Salivary glands
(3) Diaphragm
(4) Cerebral hemisphere

166. Read the following statements carefully w.r.t. humans:
(a) Hormones produce their effects on target tissue by binding to specific proteins called receptors located in the target tissue.
(b) Binding of a hormone to its receptors leads to the formation of a hormone-receptor complex.
(c) Each receptor can be activated by two or more types of hormones and hence receptors are not specific.
(d) Hormone-receptor complex formation leads to certain biochemical changes in the target tissue.
(e) Target tissues are able to regulate their own metabolism without any external support.
In the light of above statements, choose the **correct** option.
(1) Statements (a), (b) and (c) are correct
(2) Statements (a), (b) and (d) are correct
(3) Statements (b), (c) and (d) are incorrect
(4) Statements (c), (d) and (e) are incorrect

167. Choose the **odd** one w.r.t. water soluble hormones.
(1) Epinephrine (2) Insulin
(3) Prolactin (4) Cortisol

168. **Assertion (A):** Thymosins play an important role in providing humoral immunity.
Reason (R): Thymosins play a major role in the differentiation of T-lymphocytes, which also help B-cells to produce antibodies.
In the light of above statements, choose the **correct** answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

169. Glucocorticoids, particularly cortisol is not responsible to/for
(1) Stimulating GFR
(2) Maintaining the cardiovascular system
(3) Producing inflammatory reactions
(4) Stimulate RBC production

170. Choose the **incorrect** match w.r.t. hormones and their source structures/cells
(1) ANF – Atrial wall of heart
(2) Oxytocin – Trophic cells
(3) ADH – Neurosecretory cells
(4) Glucagon – α-cells of pancreas

171. Select the antagonistic pair of hormones from the options given below.
(1) LH and FSH w.r.t. gonadal activity
(2) Oxytocin and vasopressin w.r.t. water reabsorption
(3) Vasopressin and ANF w.r.t. blood pressure
(4) Secretin and CCK w.r.t. secretion of bile juice

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172. Which of the following hormones do not originate in the hypothalamic neurons?
(1) Gonadotrophins
(2) Somatostatin
(3) CRH
(4) GnRH

173. In a healthy young man, all of the following are true w.r.t. growth factors, **except**
(1) Secreted by non-endocrine tissues
(2) Also considered as hormones
(3) Essential for the regeneration of tissues
(4) Decreased level of these inhibits GHRH

174. How many of the hormones mentioned in the box below does/do not influence gonadal activity in humans?

Androgens, GnRH, LH, FSH, Melatonin, Thyroxine, Estrogen, Progesterone

Select the **correct** option.
(1) Zero (2) Three
(3) One (4) Four

175. Choose the **incorrect** option w.r.t. a hormone which is secreted from the adenohypophysis but is not secreted from pars distalis of the pituitary gland.
(1) Synthesised by the hypothalamus and is transported axonally to neurohypophysis.
(2) Interacts with membrane-bound receptors
(3) Stimulates pigmentation of skin
(4) Acts on melanin containing cells *i.e.*, melanocytes.

176. Hypersecretion of which of the following hormones in middle age results in severe disfigurement of several body parts?
(1) LH (2) GH
(3) FSH (4) MSH

177. How many of the following hormones do/does not play significant role in carbohydrate metabolism in humans?
(a) TCT
(b) T_4
(c) Cortisol
(d) Androgens
(e) Insulin
(f) Epinephrine
(g) Aldosterone
(h) Glucagon
Choose the correct option.
(1) One (2) Two
(3) Four (4) Five

178. A hormone that acts mainly on hepatocytes and stimulates glycogenolysis is secreted by a composite gland located in the region of
(1) Abdominal cavity between limbs of the C-shaped curvature of duodenum
(2) Neck, on the dorsal side of a bilobed gland
(3) Cranial cavity, on the dorsal side of brain
(4) Cranium, within a bony cavity called sella tursica

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179. Match the list I with list II and select the **correct** option.

	List I		List II
(a)	Cretinism	(i)	Hyperglycemia
(b)	Pituitary dwarfism	(ii)	Low intelligence and deaf-mutism
(c)	Diabetes insipidus	(iii)	Stunted growth due to deficiency of GH during childhood
(d)	Diabetes mellitus	(iv)	Diminished ability of kidney to conserve water

	(a)	(b)	(c)	(d)
(1)	(i)	(iii)	(iv)	(ii)
(2)	(iv)	(ii)	(i)	(iii)
(3)	(i)	(iv)	(iii)	(ii)
(4)	(ii)	(iii)	(iv)	(i)

180. Read the following facts w.r.t. the main hormone secreted by corpus luteum formed by ruptured Graafian follicle in ovary.

(a) It is a steroid hormone

(b) Regulates female sexual behaviour

(c) Supports pregnancy

(d) Acts on the mammary glands

(e) Stimulates the formation of alveoli in mammary gland

(f) Causes development of ovarian follicle

Choose the option with only **incorrect** facts.

(1) (a), (c) and (d) only

(2) (b), (d) and (e) only

(3) (a), (c), (d) and (e)

(4) (b) and (f) only

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